

Microbiology: Infectious Rheumatic Fever, Vasculitis, and Cardiomyopathy

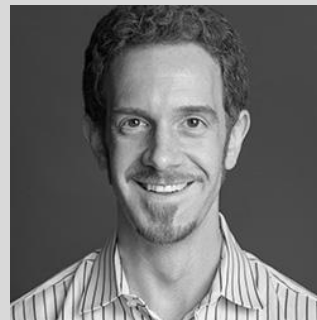
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Session Objectives

1. Describe the epidemiology associated with rheumatic fever, rocky mountain spotted fever, and Chagas disease.
2. Identify and describe the microbial pathogens associated with each disease.
3. Describe the pathogenesis and clinical presentation associated with each disease.
4. Explain the diagnosis and treatment for each disease.

Rheumatic Fever

Acute Rheumatic Fever (ARF)

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JOINT PAIN + FEVER CAN BE:
RHEUMATIC FEVER

Malaria is not the only cause of joint pain and fever. Joint pain and fever can indicate **Acute Rheumatic Fever**.

Two atipa pee n keken aye kelo arem me wang mwot kede lyeto ento twero bedo anyut me **two adunu nyo two itao.**

Acute Rheumatic Fever can damage the heart.

Two adunu nyo two itao twero balo adunu ni woko.

If your child (3-17 yrs) has joint pain and fever take them to the nearest health center for evaluation.

Cwal atini (me mwaka 3 nio 18) matye kede arem me wang mwot kede lyeto I ot yat ma cok kedi wek opim.

FREE EVALUATIONS IN LIRA REGIONAL REFERRAL HOSPITAL
Dakatal adwong me Lira
TYE ME NONO

STOP Rheumatic Heart Disease
Preventing • Treating • Reducing

0800-201900

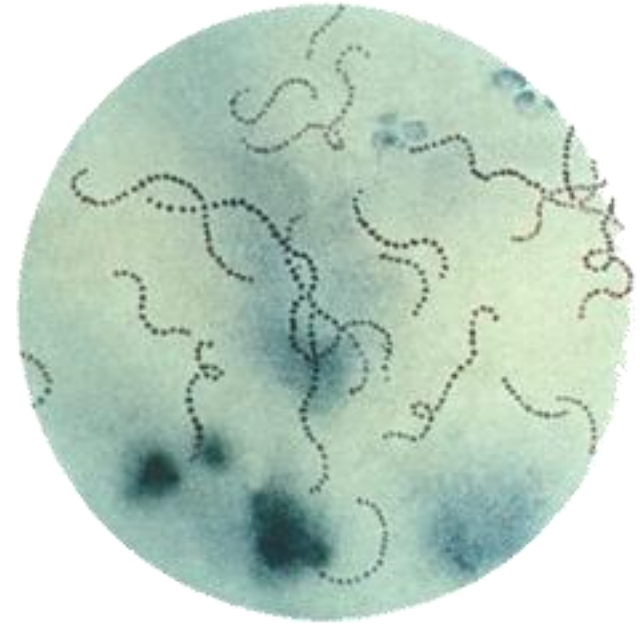
ST • P • R • H • D
Preventing • Treating • Reducing

- Multisystem disease
- Results from an autoimmune-like reaction to *Streptococcus pyogenes* infection
- Occurs around **3 weeks** after group A *Streptococcus pyogenes* pharyngitis
- Mainly a disease of **children** aged 5-15 y/o
- Higher incidence in low-resource countries

Longo, D., Fauci, A., Kasper, D., Hauser, S., Larry, J. J., Loscalzo, J., Holland, S., & Langford, C. (2025).

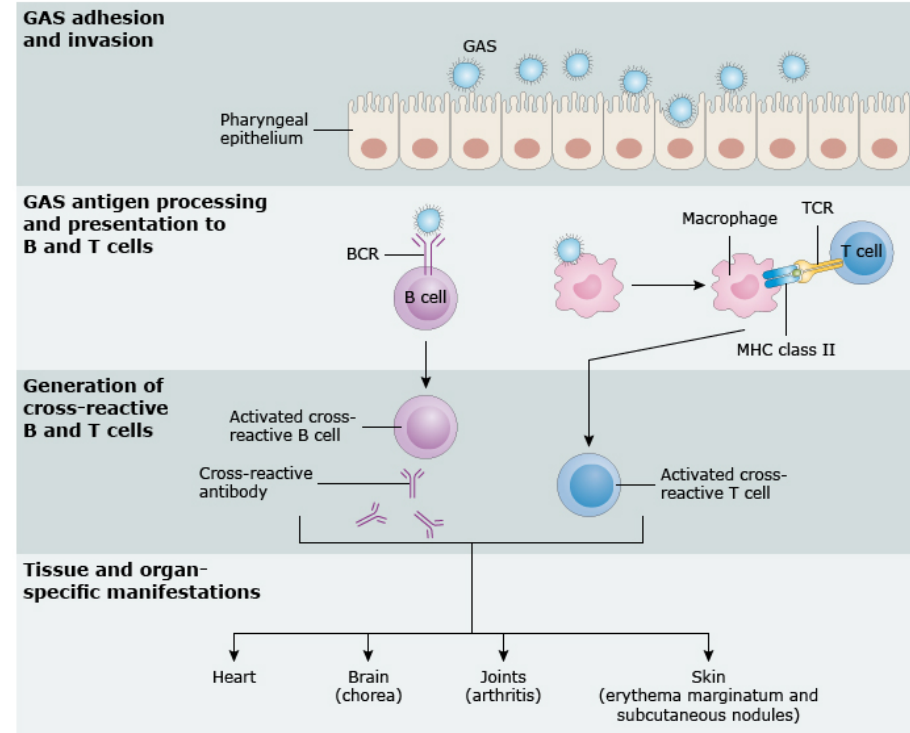
Harrison's Principles of Internal Medicine, 22nd Edition.

- ***Streptococcus pyogenes***
(Group A Streptococcus)
 - Gram positive coccus arranged in chains
 - Catalase negative
 - **β** hemolytic
 - Streptolysin S & O exotoxins
 - Susceptible to bacitracin
 - **M Protein** virulence factor
 - Hyaluronic acid capsule



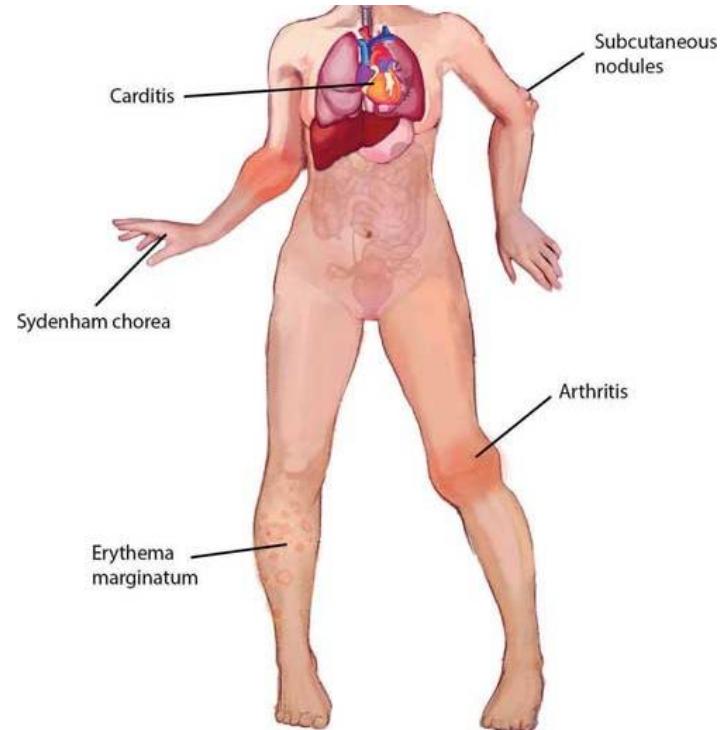
ARF: Pathophysiology

- Immune mediated
 - Type II Hypersensitivity (Ab mediated)
 - **Molecular mimicry:** Abs targeted at M protein also recognize human cardiac tissue
 - Antibodies bind to endothelial cells -> recruitment of activated lymphocytes and lysis of endothelial cells -> amplifying damage



ARF: JONES criteria

- **JONES Criteria:** rules for diagnosis of ARF
 - Required criteria and 2 major or
 - Required criteria, 1 major and 2 minor
- Required: evidence of prior group A streptococcus infection
- Major: J ❤️ NES
- Minor:
 - Fever
 - Arthralgia
 - Previous RF/RHD
 - Acute phase reactants (ESR, CRP)
 - Prolonged PR interval



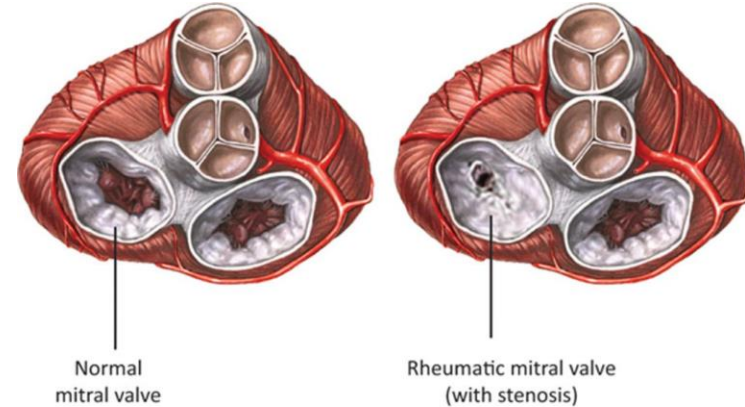
JONES Criteria

- **J = Joints**
 - Polyarthrititis (migratory)
- **N = Nodules**
- **E = Erythema marginatum**
 - Pink macules with clearing
- **S = Sydenham's chorea**
 - Rapid, involuntary muscle movements

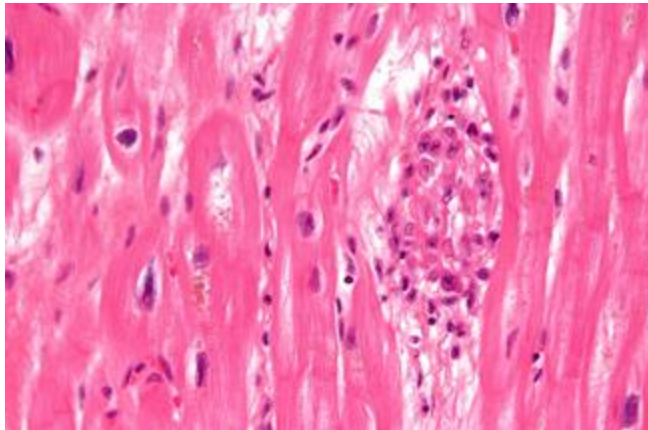


JONES Criteria

- O = ❤️
- Pancarditis
 - Endocarditis, myocarditis, pericarditis
- Can lead to cardiac valve scarring -> **stenosis**
 - Chronic RF (Rheumatic heart disease)

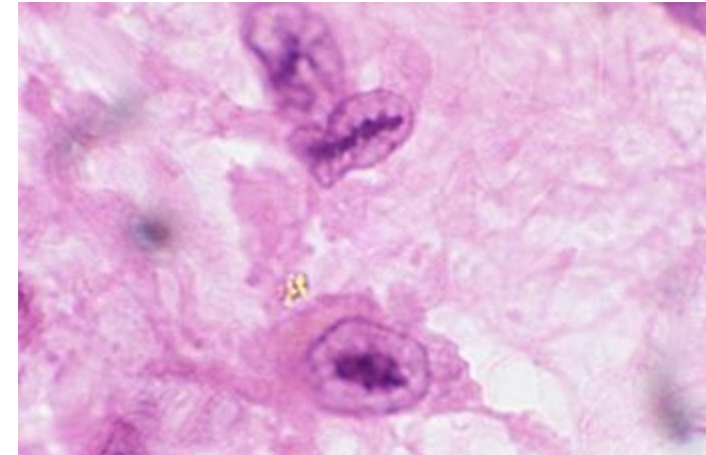


ADAM.



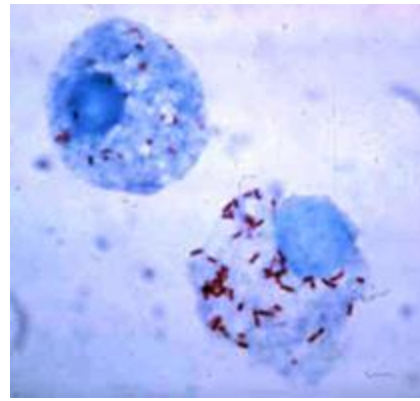
<- **Aschoff bodies**
chronic inflammation
causes granulomatous
structure

Anitschkow cells ->
Enlarged macrophages
with wavy nucleus



Bacterial Vasculitis: Rocky Mountain Spotted Fever

RMSF Etiology

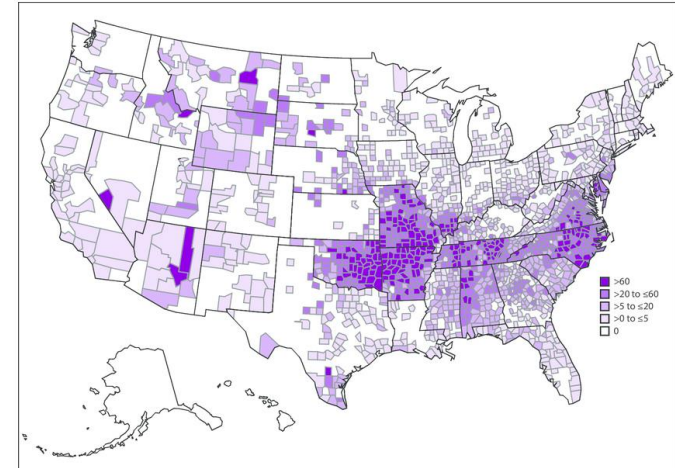


American Dog Tick (*Dermacentor variabilis*)



Incidence of Spotted Fever Rickettsiosis

Reported to CDC by County 2000 - 2013



* As reported through national surveillance, per 1,000,000 persons per year. Cases are reported by county of residence, which is not always where the infection was acquired.

+ Includes Rocky Mountain spotted fever (RMSF) and other spotted fever group rickettsioses. In 2010, the name of the reporting category changed from RMSF to spotted fever rickettsiosis.

- ***Rickettsia rickettsii***
 - Transmitted by a dog tick (*Dermacentor variabilis*)
 - Small, Gram negative coccobacillus
 - Obligate intracellular bacterium
 - Difficult to Gram-stain
 - Stain with GIEMSA
 - Bacteria replicate in cytoplasm of endothelial cells -> **vasculitis**

RMSF Clinical Presentation

- Initially, similar to a self-limiting viral illness: fever, headache, malaise, etc.
- Rash** (macules) appears first on the **wrists and ankles**, then spreads to the remainder of the extremities and trunk
 - Rash is due to **vasculitis** and inflammatory damage to the blood vessel
 - Can become necrotic vasculitis -> CNS, liver and renal damage



Diagnosis and Treatment

- Diagnosis
 - History of exposure to potentially tick-infested environment
 - Serologic test: indirect immunofluorescence assay
 - Skin biopsy
 - PCR
- Treatment: doxycycline

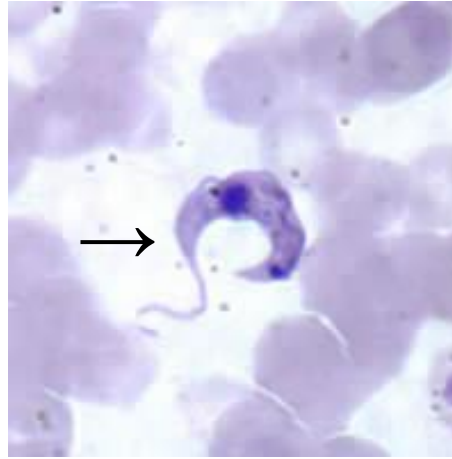


Chagas Disease (American Trypanosomiasis)

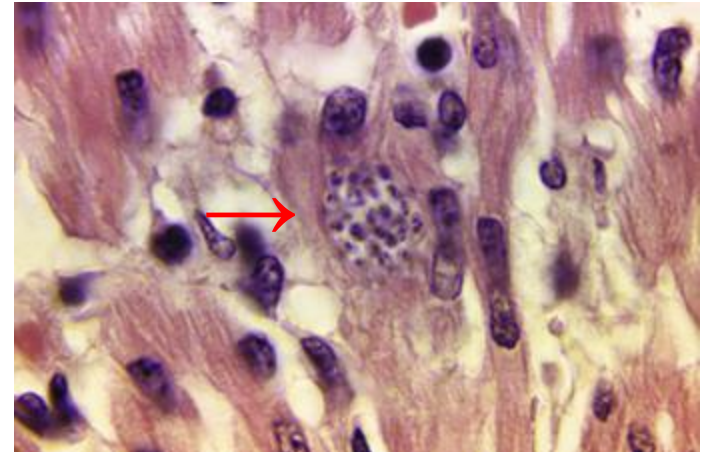
Chagas Disease Etiology

- ***Trypanosoma cruzi***

- Protozoan
- Trypomastigote form →
 - Undulating membrane
 - has flagella
- Amastigote form →
 - Intracellular
 - No flagella/undulating membrane
 - Found in tissues
 - Form pseudocysts



Murray, P. R., Rosenthal, K. S., & Pfaller, M. A. (2021). Blood and Tissue Protozoa. In *Medical microbiology* (9th ed.). Elsevier.

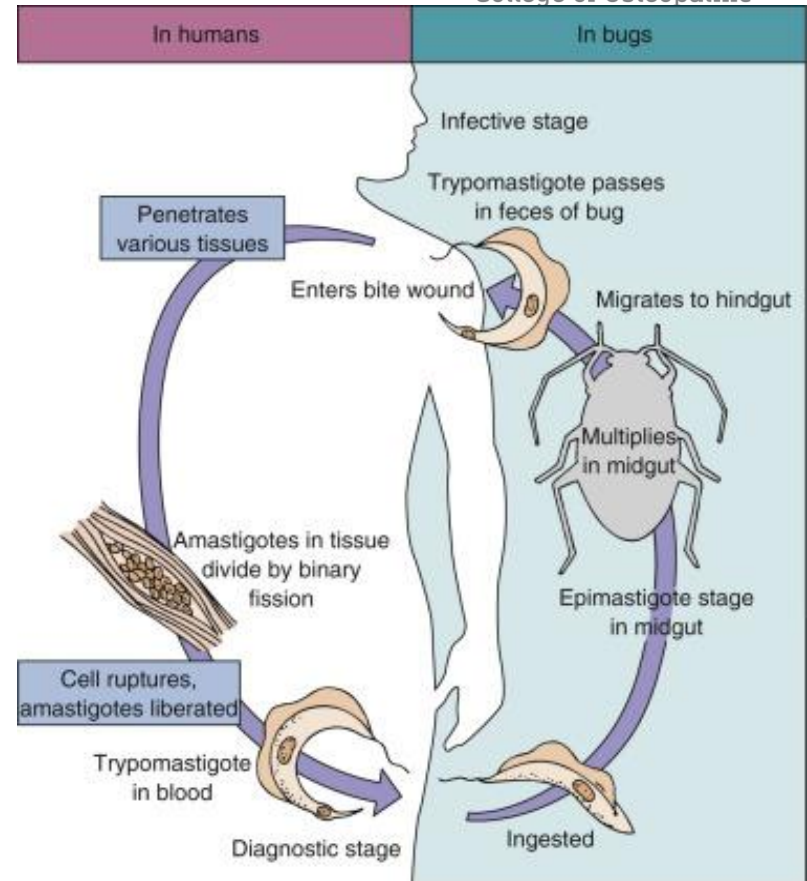


Chagas Disease: Pathophysiology

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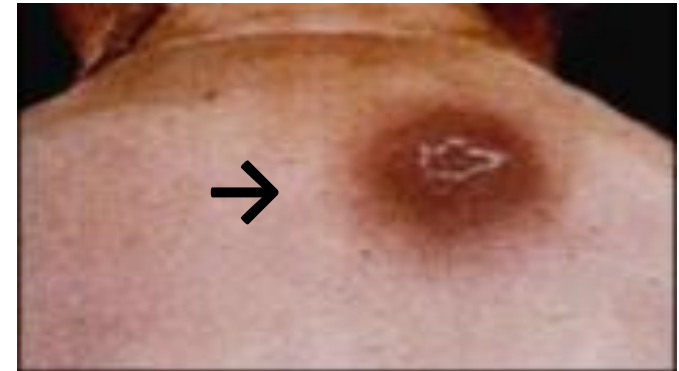
- Vector = **reduviid bug** (AKA kissing bug or assassin bug)
 - Bites around the mouth and on the face
 - Trypomastigotes in bug's feces
- Protozoa enters bloodstream/lymphatics
 - Cardiac muscle, liver, brain
 - Forms "cysts"



Chagas Disease: Clinical presentation

(Acute)

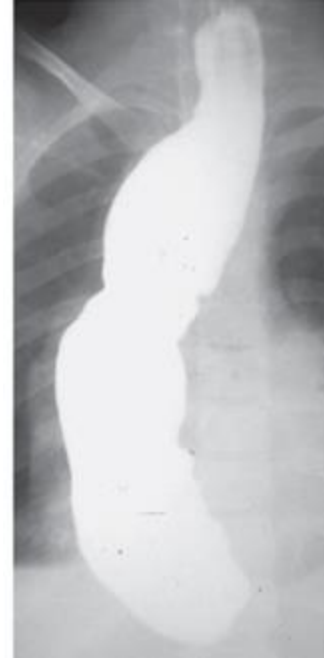
- Romaña sign: unilateral painless edema of periocular tissue (→)
- Chagoma: indurated area with local lymphadenopathy at the site of the bite (→)
- Malaise, fever, anorexia, generalized lymphadenopathy, hepatosplenomegaly
- Acute signs and symptoms usually resolve spontaneously until development into chronic chagas



Chagas Disease: Clinical Presentation

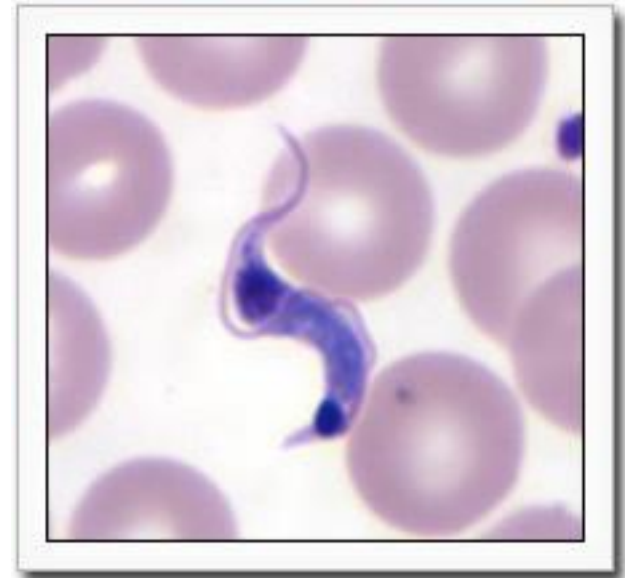
(Chronic)

- Years to decades after initial infection
- Cardiac arrhythmias
- Myocarditis
- Dilated cardiomyopathy
- Megaesophagus
- Megacolon



Chagas Disease: Diagnosis and Treatment

- Diagnosis
 - Visualize protozoa in blood, lymph node, liver, spleen, bone marrow using the Giemsa stain
 - Serological
 - PCR
- Treatment
 - Nifurtimox or benznidazole for acute cases



Summary Slide

Disease	Organism	Clinical Features	Diagnosis	Treatment
Rheumatic Fever	<i>Streptococcus pyogenes</i>	Migratory polyarthritits, carditis, subcutaneous nodules, erythema marginatum, Sydenham chorea	JONES criteria; prior strep infection	Penicillin
Rocky Mountain Spotted Fever	<i>Rickettsia rickettsii</i> Vector: <i>Dermacentor variabilis</i>	Rash that begins on the wrists and ankles then spreads to the rest of the extremities and trunk	Serology; tissue biopsy; PCR; prior tick exposure	Doxycycline
Chagas Disease	<i>Trypanosoma cruzi</i> Vector: Reduviid Bug	Acute: Chagoma, Romaña sign; Chronic: arrhythmias, Dilated cardiomyopathy, Megacolon, Megaesophagus	Visualize protozoan in blood/tissue; serology; PCR	Nifurtimox or Benznidazole

References

- Chin-Hong P. et al. 2026. Levinson's Review of Medical Microbiology & Immunology, 19th ed. McGraw-Hill Education.
- Murray, PR et. al. 2026. Medical Microbiology, 10th ed. Elsevier.
- Tao, LE et. al. 2026. First Aid for the USMLE Step 1; McGraw Hill Publication.
- UptoDate:
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 - <https://www-uptodate-com.nyit.idm.oclc.org/contents/epidemiology-clinical-manifestations-and-diagnosis-of-rocky-mountain-spotted-fever>
 - <https://www-uptodate-com.nyit.idm.oclc.org/contents/chagas-disease-chronic-trypanosoma-cruzi-infection>

THANK YOU!



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